SRINIVAS UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

MODULE 4

DEEP LEARNIG

INDIVIDUAL TASK

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INDIVIDUAL TASK

Alzheimer's Case Study Summary: Analyze the Alzheimer's video, summarize data requirements, CNN usage, model evaluation metrics, and ethical implications in ~1 page.

Introduction

Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking, and behavior. It is the most common cause of dementia among older adults. As the disease progresses, it leads to severe cognitive decline, affecting daily life and independence.

Early detection of Alzheimer’s is very important because:

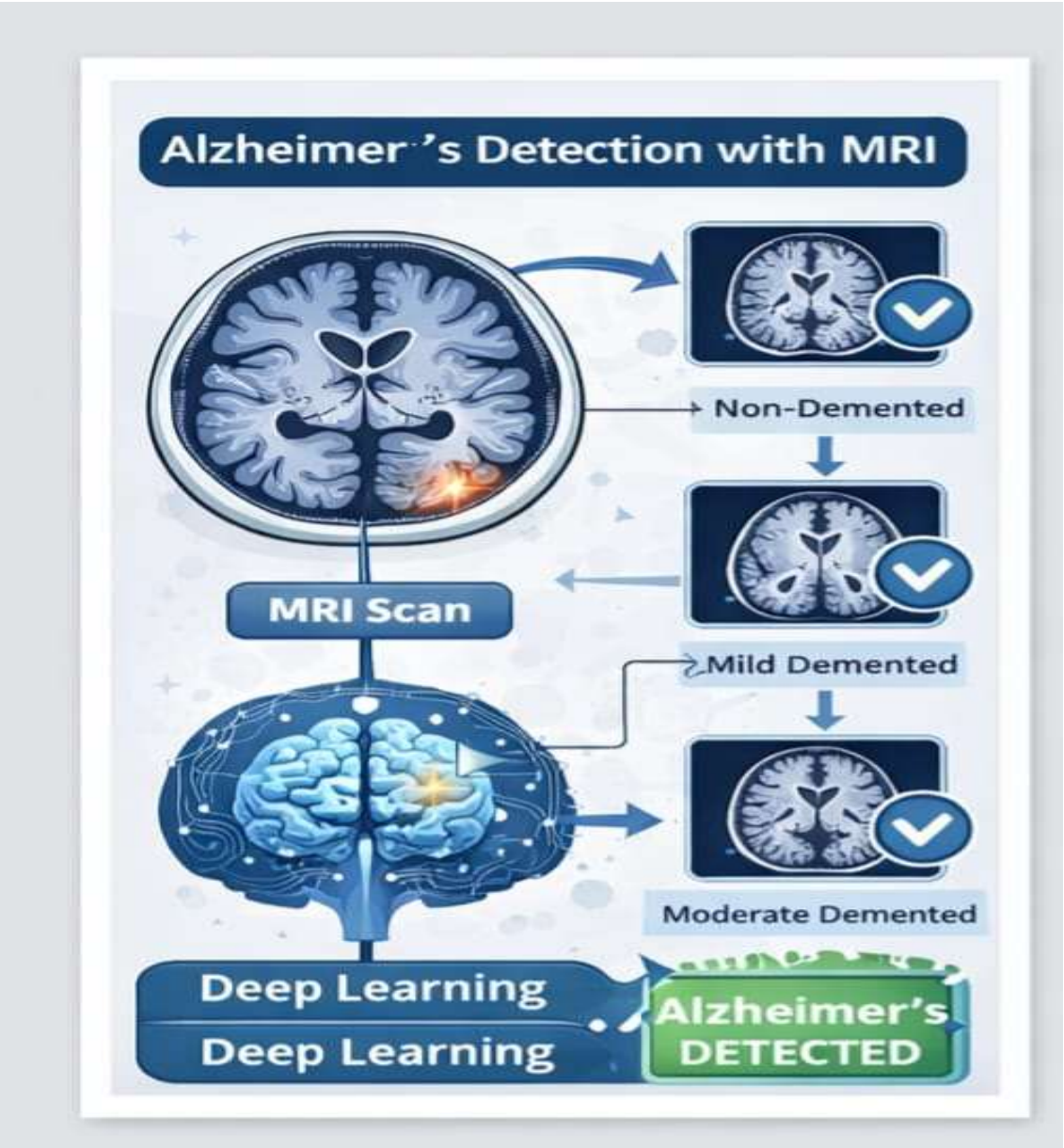
- It helps in slowing disease progression
- Improves patient care planning
- Enables early medical intervention

Traditional diagnosis methods include:

- Clinical evaluation
- MRI/CT brain scans
- Cognitive tests

However, these methods can be slow and sometimes inaccurate in early stages. This is where **deep learning** plays a major role.

Deep learning models can analyze large medical datasets and detect hidden patterns, enabling early and more accurate diagnosis.



Data Requirements

Deep learning models require high-quality and well-labeled data. For Alzheimer's detection, the following types of data are used:

1. Medical Imaging Data

- MRI (Magnetic Resonance Imaging)
- PET (Positron Emission Tomography)
- CT scans

MRI is most commonly used because it shows brain structure clearly.

2. Dataset Examples

- ADNI (Alzheimer's Disease Neuroimaging Initiative)
- OASIS Brain Dataset

3. Data Preprocessing

Before training the model, data must be cleaned and prepared:

- Image resizing
- Noise removal
- Normalization
- Data augmentation (rotation, flipping)

4. Labeling

Data is labeled into categories such as:

- Non-Demented
- Mild Demented
- Moderate Demented
- Severe Demented

Challenges

- Limited dataset availability
- Imbalanced classes
- High cost of medical imaging

CNN Usage in Alzheimer's Detection

Convolutional Neural Networks (CNNs) are widely used for image-based tasks like Alzheimer's detection.

Why CNN?

- Automatically extracts features from images
- Detects patterns in brain structure
- Reduces need for manual feature extraction

CNN Architecture

A typical CNN consists of:

1. **Input Layer** – Brain scan images
2. **Convolution Layer** – Extracts features
3. **Pooling Layer** – Reduces size
4. **Fully Connected Layer** – Classification
5. **Output Layer** – Disease category

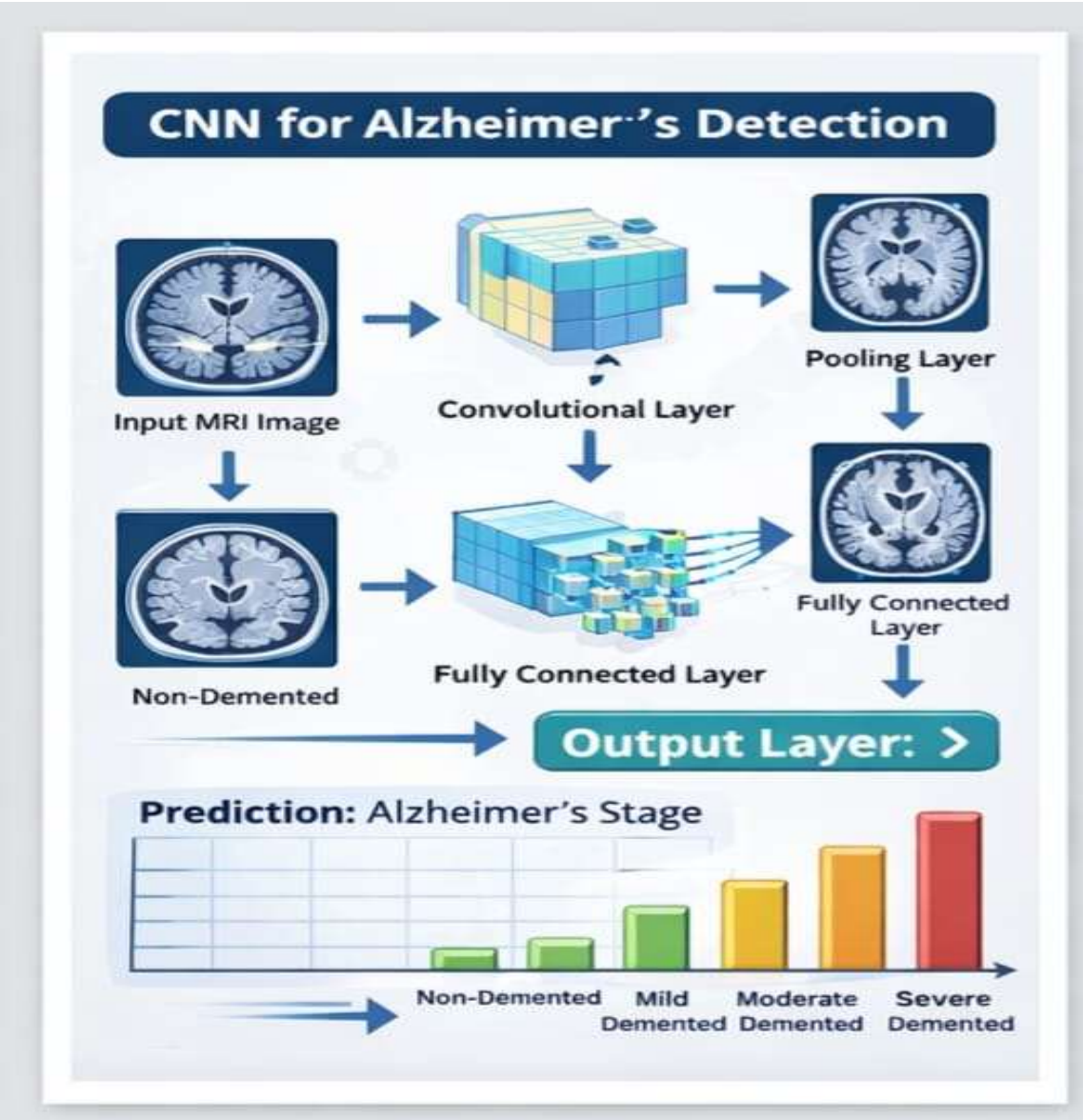
Working Process

1. Input MRI image
2. CNN extracts important features
3. Model learns patterns of Alzheimer's
4. Predicts disease stage

Popular Models

- VGG16
- ResNet
- AlexNet

CNN models can achieve high accuracy in detecting early-stage Alzheimer's.



Model Evaluation Metrics

To measure the performance of the deep learning model, several evaluation metrics are used:

1. Accuracy

Percentage of correct predictions
Formula:
Accuracy = (Correct Predictions / Total Predictions)

2. Precision

Measures how many predicted positives are actually correct

3. Recall (Sensitivity)

Measures how many actual positives are detected

4. F1 Score

Balance between precision and recall

5. Confusion Matrix

Shows:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Importance

- Helps compare different models
- Ensures reliability in medical diagnosis
- Reduces false predictions

Ethical Implications

Using deep learning in healthcare raises several ethical concerns:

1. Data Privacy

- Patient data must be protected
- Secure storage and sharing required

2. Bias in Data

- Models may perform poorly on underrepresented groups
- Leads to unfair predictions

3. Reliability

- Incorrect diagnosis can harm patients
- Models must be highly accurate

4. Transparency

- Deep learning models are often “black boxes”
- Hard to explain decisions

5. Responsibility

- Who is responsible if the model fails?
    - Doctor?
    - Developer?

Ethical AI is necessary to ensure safe and fair usage in healthcare.

Conclusion

Deep learning has revolutionized the detection of Alzheimer’s disease. By using CNNs and medical imaging data, early diagnosis has become more accurate and efficient.

Key Points

- CNNs are powerful for image-based diagnosis
- Data quality is critical
- Evaluation metrics ensure model performance
- Ethical considerations must be addressed

Future Scope

- Integration with wearable devices
- Real-time diagnosis systems
- Improved explainable AI models
- Larger and diverse datasets